Worksheet 2.1 — Power, Constant & Sum Rules

Grade 12 Calculus and Vectors (MCV4U) · Unit 2: Derivative Rules

The Big Ideas

$\frac{d}{dx}x^n = nx^{n-1}$, with the constant-multiple and sum/difference rules, differentiates any polynomial. Rewrite radicals and reciprocals as powers first.

- Power: $\frac{d}{dx}x^n = nx^{n-1}$; constant: $\frac{d}{dx}c = 0$.
- Constant multiple and sum/difference rules.
- Rewrite $\sqrt{x} = x^{1/2}$, $\frac{1}{x^k} = x^{-k}$.

The Big Picture

The power rule

For $f(x) = x^n$ the derivative is $f'(x) = nx^{n-1}$ the workhorse of differentiation.

Constants and sums

A constant multiple factors out, $\frac{d}{dx}[cf] = cf'$, and derivatives add term by term, $\frac{d}{dx}[f+g] = f' + g'$.

Together

These rules differentiate any polynomial instantly, one term at a time.

Worked Examples

Example 1 — Power rule

Differentiate $f(x) = x^5$.

Solution. **Step 1 (identify n):** the power rule says $\frac{d}{dx}x^n = nx^{n-1}$; here $n = 5$.

Step 2 (apply it): bring the 5 down and subtract 1 from the exponent: $5x^{5-1}$.

Conclusion: $f'(x) = 5x^4$.

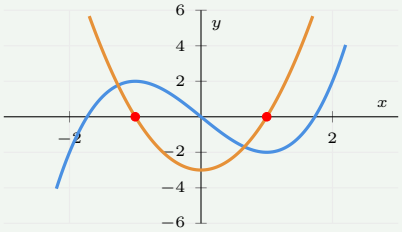
Example 2 — Whole polynomial

Differentiate $f(x) = 3x^4 - 2x^2 + 7$.

Solution. **Step 1 (differentiate term by term):** $\frac{d}{dx}3x^4 = 12x^3$, $\frac{d}{dx}(-2x^2) = -4x$, $\frac{d}{dx}7 = 0$ (constants vanish).

Step 2 (combine): $12x^3 - 4x + 0$.

Conclusion: $f'(x) = 12x^3 - 4x$. f and f' for $x^3 - 3x$ illustrate the idea — f' is zero at the turning points:



Example 3 — Negative & fractional

Differentiate $f(x) = \sqrt{x} + \frac{1}{x^2}$.

Solution. **Step 1 (rewrite as powers):** $\sqrt{x} = x^{1/2}$ and $\frac{1}{x^2} = x^{-2}$.

Step 2 (apply the power rule to each term): $\frac{d}{dx}x^{1/2} = \frac{1}{2}x^{-1/2}$ and $\frac{d}{dx}x^{-2} = -2x^{-3}$.

Step 3 (rewrite back in radical/fraction form): $\frac{1}{2}x^{-1/2} = \frac{1}{2\sqrt{x}}$ and $-2x^{-3} = -\frac{2}{x^3}$.

Conclusion: $f'(x) = \frac{1}{2\sqrt{x}} - \frac{2}{x^3}$.

Example 4 — Slope at a point

Slope of the tangent to $f(x) = x^3 - 3x$ at $x = 2$.

Solution. **Step 1 (differentiate):** $f'(x) = 3x^2 - 3$.

Step 2 (substitute $x = 2$): $f'(2) = 3(2)^2 - 3 = 3(4) - 3$.

Conclusion: $= 9$.

Example 5 — Horizontal tangent

Where does $f(x) = x^3 - 3x$ have a horizontal tangent?

Solution. **Step 1 (differentiate):** $f'(x) = 3x^2 - 3$.

Step 2 (set the slope to 0, since a horizontal tangent has slope 0): $3x^2 - 3 = 0$.

Step 3 (solve): $3x^2 = 3 \Rightarrow x^2 = 1 \Rightarrow x = \pm 1$.

Conclusion: horizontal tangents at $x = 1$ and $x = -1$.

Example 6 — Power rule

Differentiate $f(x) = x^7$.

Solution. **Step 1 (identify n):** $n = 7$.

Step 2 (apply the power rule): $7x^{7-1}$.

Conclusion: $f'(x) = 7x^6$.

Example 7 — Polynomial

Differentiate $f(x) = 4x^3 - 6x^2 + x - 9$.

Solution. **Step 1 (differentiate term by term):** $\frac{d}{dx}4x^3 = 12x^2$, $\frac{d}{dx}(-6x^2) = -12x$, $\frac{d}{dx}x = 1$, $\frac{d}{dx}(-9) = 0$.

Step 2 (combine): $12x^2 - 12x + 1 + 0$.

Conclusion: $f'(x) = 12x^2 - 12x + 1$.

Example 8 — Reciprocal power

Differentiate $f(x) = \frac{1}{x^3}$.

Solution. **Step 1 (rewrite as a power):** $\frac{1}{x^3} = x^{-3}$.

Step 2 (apply the power rule): $-3x^{-3-1} = -3x^{-4}$.

Step 3 (rewrite back as a fraction): $-3x^{-4} = -\frac{3}{x^4}$.

Conclusion: $f'(x) = -\frac{3}{x^4}$.

Example 9 — Evaluate a derivative

Find $f'(1)$ for $f(x) = x^4 - 2x$.

Solution. **Step 1 (differentiate):** $f'(x) = 4x^3 - 2$.

Step 2 (substitute $x = 1$): $f'(1) = 4(1)^3 - 2 = 4 - 2$.

Conclusion: $= 2$.

Your Turn — Show Your Work

For each question, show all steps clearly in the space provided.

Question 1

Differentiate $f(x) = x^9$.

Question 2

Differentiate $f(x) = 2x^5 - 3x^3 + 4$.

Question 3

Differentiate $f(x) = \dfrac{1}{x^2}$.

Question 4

Differentiate $f(x) = \sqrt{x}$.

Question 5

Find $f'(2)$ for $f(x) = x^3 - x$.

Question 6

Where does $f(x) = x^2 - 6x$ have a horizontal tangent?

Question 7

Differentiate $f(x) = 5x^4 - x$.

Question 8

Differentiate $f(x) = x^{1/3}$.

Question 9

Find $f'(0)$ for $f(x) = x^3 + 2x + 1$.

Question 10

Differentiate $f(x) = \dfrac{3}{x}$.

Question 11

Where is the tangent to $f(x) = x^3 - 12x$ horizontal?

Question 12

Differentiate $f(x) = 6\sqrt{x}$.

Question 13 — Challenge

Find the equation of the tangent to $f(x) = x^2 - 2x$ at $x = 3$.

Question Recap & Answer Key

Question Recap

1. Differentiate $f(x) = x^9$.
2. Differentiate $f(x) = 2x^5 - 3x^3 + 4$.
3. Differentiate $f(x) = \dfrac{1}{x^2}$.
4. Differentiate $f(x) = \sqrt{x}$.
5. Find $f'(2)$ for $f(x) = x^3 - x$.
6. Where does $f(x) = x^2 - 6x$ have a horizontal tangent?
7. Differentiate $f(x) = 5x^4 - x$.
8. Differentiate $f(x) = x^{1/3}$.
9. Find $f'(0)$ for $f(x) = x^3 + 2x + 1$.
10. Differentiate $f(x) = \dfrac{3}{x}$.
11. Where is the tangent to $f(x) = x^3 - 12x$ horizontal?
12. Differentiate $f(x) = 6\sqrt{x}$.
13. Find the equation of the tangent to $f(x) = x^2 - 2x$ at $x = 3$.

✓ Answer Key

1. $9x^8$
2. $10x^4 - 9x^2$
3. $-\frac{2}{x^3}$
4. $\frac{1}{2\sqrt{x}}$
5. $3(4) - 1 = 11$
6. $x = 3$
7. $20x^3 - 1$
8. $\frac{1}{3}x^{-2/3}$
9. 2
10. $-\frac{3}{x^2}$
11. $x = \pm 2$
12. $\frac{3}{\sqrt{x}}$
13. $f'(3) = 4$, point $(3, 3)$: $y = 4x - 9$